

title: "rv32im_core — Hardware IP Specification"
revision: "0.2"
date: "2026-08-29"
status: "Released"

Field	Value
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IP Name	rv32im_core
Revision	0.2
Date	2026-08-29
Status	Released
Tool Flow	Claude Code RTL Gen Flow
Gate Status	Gate 5 ✓ (All phases complete)

Quick Reference

Module Hierarchy

```
rv32im_core (top)
├── rv32im_if_stage    ─ Instruction Fetch stage
│   └── (PC register, branch/trap redirect)
├── rv32im_id_stage    ─ Instruction Decode stage
│   └── rv32im_decoder ─ Opcode decode + control signals
│       ├── rv32im_imm_gen ─ Immediate value generator
│       └── rv32im_regfile ─ 32x32 register file (x0-x31)
├── rv32im_ex_stage    ─ Execute stage
│   ├── rv32im_alu      ─ ALU (RV32I ops)
│   ├── rv32im_muldiv   ─ Multinlv/Divide unit (PR_M_EXT_EN=1)
│   │   ├── rv32im_mult ─ Multiplier
│   │   └── rv32im_div  ─ Divider (iterative)
│   └── rv32im_branch_unit ─ Branch condition + target
├── rv32im_csr_file    ─ CSR register bank (PR_CSR_EN=1)
├── rv32im_mem_stage    ─ Memory Access stage
│   └── rv32im_lsu      ─ Load/Store Unit, byte-enable gen
├── rv32im_trap_ctrl   ─ Trap & interrupt controller (PR_TR0_EN=1)
└── rv32im_hazard_ctrl ─ Pipeline hazard detection & forwarding
```

Parameter Summary

Parameter	Type	Default	Description
----- ----- ----- -----			

PR_BOOT_ADDR	logic [31:0]	32'h8000_0000	PC reset / boot address
PR_MTVEC_RESET	logic [31:0]	32'h0000_0000	mtvec reset value
PR_HART_ID	logic [31:0]	32'h0	Hardware thread ID (mhartid CS)
PR_M_EXT_EN	bit	1	Enable RV32M (MUL/DIV) extensi
PR_MULT_IMPL	int unsigned	0	Multiplier implementation (0=s
PR_DIV_IMPL	int unsigned	0	Divider implementation (0=iter
PR_CSR_EN	bit	1	Enable CSR register file
PR_IRQ_EN	bit	1	Enable interrupt/trap handling
PR_COUNTER_EN	bit	1	Enable mcycle/minstret counter
PR_MTVEC_VEC_EN	bit	0	Vectored trap mode (0=direct)
PR_FWD_EN	bit	1	Enable EX→EX and MEM→EX forwar
PR_RF_RESET_EN	bit	1	Reset register file to zero
PR_RF_IMPL	int unsigned	0	Reg-file implementation (0=FF-
PR_TRACE_EN	bit	0	Enable commit trace port
PR_BUS_OUTSTANDING	int unsigned	1	Max outstanding bus requests

Feature Matrix

Feature	Description	Enabled	Parameter
----- ----- ----- -----			
F01	RV32I base ISA (40 instruction ✓	—	—
F02	RV32M multiply/divide extensio ✓	PR_M_EXT_EN=1	—
F03	FENCE.I instruction support ✓	—	—

F04	WFI instruction support	✓	—
F05	Configurable boot address	✓	PR_BOOT_ADDR
F06	Load-use stall (1-cycle penalt	✓	—
F07	EX→EX and MEM→EX data forwardi	✓	PR_FWD_EN=1
F08	Branch resolution + pipeline f	✓	—
F09	32×32 register file (x0 hardwi	✓	PR_RF_IMPL=0
F10	Load/Store byte-enable generat	✓	—
F11	CSR register bank	✓	PR_CSR_EN=1
F12	Machine-mode trap handling (sy	✓	PR_IRQ_EN=1
F13	Machine-mode interrupt handlin	✓	PR_IRQ_EN=1
F14	Instruction fetch bus protocol	✓	—
F15	Data memory bus protocol	✓	—
F16	Bus error handling	✓	—
F17	Performance counters (mcycle,	✓	PR_COUNTER_EN=1
F18	Vectored trap mode	✗	PR_MTVEC_VEC_EN=0
F19	Commit trace port	✗	PR_TRACE_EN=0

01 — Overview

1.1 Introduction

rv32im_core is a 32-bit RISC-V in-order scalar processor core implementing the RV32IM instruction set architecture. It implements a classic 5-stage pipeline (IF → ID → EX → MEM → WB) with full data hazard handling via forwarding and stalling. The core presents a Harvard-style memory interface: a dedicated instruction bus (I-Bus) and a dedicated data bus (D-Bus), both using a simple valid/ready handshake protocol.

The core targets embedded MCU-class applications and is synthesisable to GF180MCU and other standard-cell technologies. All optional features (M-extension, CSR, interrupts, trace) are parameterisable at elaboration time.

1.2 Key Features

- RV32I base ISA: 40 instructions (37 computational + FENCE + ECALL + EBREAK)
- RV32M extension: MUL, MULH, MULHSU, MULHU, DIV, DIVU, REM, REMU
- 5-stage in-order pipeline: IF / ID / EX / MEM / WB
- Single-issue, no branch prediction (flush on taken branch)
- Load-use hazard detection with 1-cycle stall insertion
- EX→EX and MEM→EX data forwarding (configurable)
- Machine-mode CSR: mstatus, mie, mip, mtvec, mepc, mcause, mtval, mscratch, mhartid, mcycle, minstret
- Machine-mode interrupts: software (MSIP), timer (MTIP), external (MEIP)
- Synchronous exception handling: illegal instruction, ECALL, EBREAK, load/store misalign, bus error
- Configurable boot address via PR_BOOT_ADDR
- Harvard bus interface with valid/ready handshake

1.3 Top-Level Parameters

See Quick Reference → Parameter Summary above.

1.4 Module Hierarchy

See Quick Reference → Module Hierarchy above.

02 — Port List

Clock & Reset

Port	Direction	Width	Description
i_clk_core	input	1	Core clock (all FFs on rising
i_resetrn_core	input	1	Active-low synchronous reset

Instruction Bus (I-Bus)

Port	Direction	Width	Description
o_imem_req_valid	output	1	Fetch request valid
i_imem_req_ready	input	1	Fetch request accepted by memo
o_imem_req_addr	output	32	Fetch address (word-aligned)
i_imem_rsp_valid	input	1	Instruction data valid
i_imem_rsp_rdata	input	32	Instruction word
i_imem_rsp_err	input	1	Instruction fetch bus error

Data Bus (D-Bus)

Port	Direction	Width	Description
o_dmem_req_valid	output	1	Data request valid
i_dmem_req_ready	input	1	Data request accepted
o_dmem_req_addr	output	32	Data address
o_dmem_req_we	output	1	Write enable (1=write, 0=read)
o_dmem_req_be	output	4	Byte enable (one bit per byte)
o_dmem_req_wdata	output	32	Write data
i_dmem_rsp_valid	input	1	Data response valid
i_dmem_rsp_rdata	input	32	Read data
i_dmem_rsp_err	input	1	Data bus error

Interrupt / IRQ

Port	Direction	Width	Description
i_irq_sw	input	1	Software interrupt (MSIP)
i_irq_timer	input	1	Timer interrupt (MTIP)
i_irq_ext	input	1	External interrupt (MEIP)

Trace / Debug (PR_TRACE_EN=0 in this config — ports present but driven 0)

Port	Direction	Width	Description
o_trace_valid	output	1	Instruction committed this cyc
o_trace_pc	output	32	PC of committed instruction
o_trace_instr	output	32	Raw instruction word
o_trace_rd_wen	output	1	Register write enable
o_trace_rd_addr	output	5	Destination register index
o_trace_rd_wdata	output	32	Value written to register

03 — Clock & Reset

Clock Domain

Domain	Signal	Description
clk0	i_clk_core	Single clock domain — all FFs

The core is fully synchronous. There are no asynchronous paths or multi-clock-domain crossings.

Reset Strategy

- Type: Active-low synchronous reset (i_resetrn_core)
- Assertion: Held low for at least 1 clock cycle
- Effect on pipeline: All pipeline registers, CSRs, and control state cleared to zero
- Exceptions: Register file reset depends on PR_RF_RESET_EN (= 1 in this config → x0-x31 reset to 0)

Boot Sequence

On reset de-assertion:

- PC is loaded with PR_BOOT_ADDR = 32'h8000_0000
- I-Bus begins fetching from that address on the next rising edge
- CSRs reset to their defined power-on values (mstatus.MIE = 0, mtvec = PR_MTVVEC_RESET = 32'h0000_0000)
- Pipeline fills over 4 cycles before first instruction retires

04 — Microarchitecture

Pipeline Overview

rv32im_core implements a 5-stage in-order scalar pipeline:



- Issue width: 1 (single-issue)
- Execution model: In-order; no out-of-order issue, no speculative execution beyond branch flush
- Structural hazards: None — each pipeline resource is single-use per stage

Stage Description

Stage	Module	Function
IF	rv32im_if_stage	PC generation, I-Bus request,
ID	rv32im_id_stage	Decode, register read, immEDIA
EX	rv32im_ex_stage	ALU / MUL-DIV / branch resolut
MEM	rv32im_mem_stage	D-Bus load/store, byte-enable,
WB	(within top core)	Write-back to register file

Supporting modules:

Module	Function
rv32im_hazard_ctrl	Load-use stall detection, forw
rv32im_trap_ctrl	Synchronous exception + async
rv32im_csr_file	Machine-mode CSR bank (mstatus

Hazard Handling

Data hazards — forwarding (PR_FWD_EN=1):

- EX→EX: result of instruction N−1 forwarded to EX of instruction N
- MEM→EX: result of instruction N−2 forwarded before register-file WB

Load-use hazard (F06):

- Detected in ID when rs1/rs2 match load destination in EX
- Pipeline inserts 1 stall cycle (ID/EX bubble, IF/ID held)
- No multi-cycle stall needed (single outstanding bus request, PR_BUS_OUTSTANDING=1)

Control hazards (F08):

- Branch condition resolved in EX stage
- On taken branch: flush IF and ID (2-cycle branch penalty)
- On not-taken branch: no flush (instructions already in IF/ID are correct)

Instruction Execution Flow

```
IF: PC → I-Bus request → instruction word latched
ID: Instruction decode (rv32im decoder)
    Immediate generation (rv32im imm gen)
    Register file read (rv32im readfile, rs1/rs2)
    Hazard check (rv32im hazard ctrl)
EX: ALU operation / MUL-DIV execution / CSR read-modify-write
    Branch condition evaluation (rv32im_branch_unit)
    Forwarded operands applied
MEM: D-Bus request for load/store (rv32im_lsu)
    Byte-enable computation
    Load data alignment (byte/halfword sign/zero extend)
```

WR: Write rd via register-file write port
Trace output if PR_TRACE_EN=1

05 — CSR / Register Map

(Active because PR_CSR_EN=1)

Implemented CSRs

Address	Name	Access	Reset	Description
0x300	mstatus	RW	0x0000_0000	Machine status (MIE, MPIE, MPP)
0x304	mie	RW	0x0000_0000	Machine interrupt enable (MSIE)
0x305	mtvec	RW	PR_MTVEC_RESET	Trap vector base (MODE=0: direct)
0x340	mscratch	RW	0x0000_0000	Scratch register for trap handling
0x341	mepc	RW	0x0000_0000	Exception program counter
0x342	mcause	RW	0x0000_0000	Trap cause code (bit[31]: interrupt)
0x343	mtval	RW	0x0000_0000	Trap value (fault address / instruction address)
0x344	mip	RO	—	Machine interrupt pending (read-only)
0xF14	mhartid	RO	PR_HART_ID	Hardware thread ID
0xB00	mcycle	RW	0x0000_0000	Cycle counter, lower 32 bits
0xB80	mcycleh	RW	0x0000_0000	Cycle counter, upper 32 bits
0xB02	minstret	RW	0x0000_0000	Retired instruction counter, lower 32 bits
0xB82	minstreth	RW	0x0000_0000	Retired instruction counter, upper 32 bits

Vectored Mode

PR_MTVEC_VEC_EN=0 — trap mode is direct (MODE=0). All traps jump to mtvec.BASE. Vectored mode (MODE=1) is not enabled in this configuration.

06 — Functional Description

6.1 Instruction Fetch (IF)

The IF stage holds the program counter (PC) register and drives the instruction bus. On each cycle where o_imem_req_valid is asserted, it presents the next PC on o_imem_req_addr. Backpressure from memory is handled via i_imem_req_ready; the PC does not advance when the request is not accepted.

On reset: $PC \leftarrow PR_BOOT_ADDR = 32'h8000_0000$.

On taken branch (resolved in EX): $PC \leftarrow$ branch target, IF and ID pipeline registers are flushed.

On trap entry: $PC \leftarrow$ mtvec.BASE.

On MRET: $PC \leftarrow$ mepc.

6.2 Instruction Decode (ID)

The ID stage decodes the 32-bit instruction word from IF. rv32im_decoder generates all control signals (ALU op, memory op, CSR op, branch type, register write enables). rv32im_imm_gen extracts and sign-extends the immediate field (I/S/B/U/J types). rv32im_regfile performs two concurrent register reads (rs1, rs2).

Hazard detection runs in ID: if the instruction in EX is a load and its destination register matches rs1 or rs2 of the incoming instruction, a stall bubble is inserted.

6.3 Execute (EX)

The EX stage is the computational core. Forwarded operands from EX/MEM are multiplexed in before the operation. The ALU handles all RV32I arithmetic, logical, shift, and comparison operations. The rv32im_muldiv unit handles RV32M operations (active because PR_M_EXT_EN=1).

rv32im_branch_unit computes the branch condition and target address. On a taken branch,

a flush signal propagates to IF and ID.

CSR operations (CSRRW/CSRRS/CSRRC/CSRRWI/CSRRSI/CSRRCI) are handled by `rv32im_csr_file` in EX, with the read value forwarded as the EX result.

6.4 Memory Access (MEM)

`rv32im_lsu` in MEM translates load/store operations into D-Bus transactions. It computes the byte-enable mask from the access width (byte/halfword/word) and alignment. On load, it waits for `i_dmem_rsp_valid`, then sign- or zero-extends the returned data to 32 bits based on the access type (LB, LBU, LH, LHU, LW).

Bus errors on either bus (`i_imem_rsp_err`, `i_dmem_rsp_err`) trigger the corresponding synchronous exception via `rv32im_trap_ctrl`.

6.5 Write-Back (WB)

The WB stage writes the result (from ALU, load data, or CSR read) to the register file via `rv32im_regfile`'s write port. `x0` writes are suppressed in hardware. If `PR_TRACE_EN` were enabled, the commit trace outputs would also be driven here.

6.6 Hazard Control

`rv32im_hazard_ctrl` implements two mechanisms:

- Load-use stall: Detects when EX instruction is a load and ID instruction reads the same register. Stalls IF and ID for one cycle, inserts a NOP bubble into EX.
- Forwarding (`PR_FWD_EN=1`): Selects forwarded values from EX-stage output or MEM-stage output when the destination register of a prior instruction matches `rs1` or `rs2` of the current EX instruction.

6.7 Trap & Interrupt Handling (`PR_IRQ_EN=1`)

`rv32im_trap_ctrl` arbitrates between synchronous exceptions (illegal instruction, ECALL, EBREAK, misaligned access, bus error) and asynchronous interrupts (MSIP, MTIP, MEIP).

Exception entry:

- `mepc` $\leftarrow$ PC of excepting instruction (or next PC for async interrupts)
- `mcause` $\leftarrow$ cause code
- `mtval` $\leftarrow$ fault address or instruction word (as applicable)
- `mstatus.MPIE` $\leftarrow$ `mstatus.MIE`; `mstatus.MIE` $\leftarrow$ 0
- `PC` $\leftarrow$ `mtvec.BASE` (direct mode)

MRET:

- `mstatus.MIE` $\leftarrow$ `mstatus.MPIE`; `mstatus.MPIE` $\leftarrow$ 1
- `PC` $\leftarrow$ `mepc`

Interrupts are only taken when `mstatus.MIE=1` and the corresponding enable bit in `mie` is set.

07 — Verification Summary

Simulation Results

Metric	Value
----- -----	
Total TC	24
PASS	24
FAIL	0
SVA violations	0
Timeout	0
Simulator	VCS X-2025.06

Test Coverage

REQ-ID Group	Total	Covered by TC	%
----- ----- -----			
Functional	18	13	72%
Constraint	5	0	0%

Total	23	13	57%
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Mutation Testing

REQ-ID	Module	Mutants Tested	Killed	Score	Status
REQ-F06	rv32im_hazard_ctrl	20	14	70%	Below threshold
REQ-F07	rv32im_hazard_ctrl	20	14	70%	Below threshold
REQ-F08	rv32im_branch_unit	20	6	30%	Below threshold
REQ-F09	rv32im_regfile	N/A	—	N/A	No mutants tested
REQ-F12	rv32im_trap_ctrl	20	9	45%	Below threshold
REQ-F13	rv32im_trap_ctrl	20	9	45%	Below threshold

Threshold: 85%. Scores below threshold lock sign-off for that REQ-ID.

To improve F08 (30%): add test cases covering branch edge cases (boundary conditions, consecutive branches). F06/F07 (70%): add forwarding corner cases. F12/F13 (45%): add exception priority scenarios.

Sign-Off Summary

Status	REQ-IDs	Count
Signed-off	F01, F02, F09, F10, F11, F14,	8
Locked (mutation < 85%)	F06, F07, F08, F12, F13	5
Locked (no TC)	F05, F17, F18, C01, C02, C03,	8
Locked (no SVA)	F03, F04	2
Total REQ		23

08 — Synthesis Report

Summary

Metric	Value
PDK	Yosys generic gate library
PDK Note	GF180MCU liberty stub has no s
Total Cells	16,669
Sequential (FFs)	2,213
MUX cells	2,853
AND variants	5,813
OR variants	4,197
XOR variants	804
Inverters	773
Check problems	0
Lint (Verilator)	PASS (0 warnings)
Elaboration (VCS)	PASS (0 errors, 0 warnings)

Cell Breakdown (Yosys generic primitives)

Cell	Count
\$_DFF_PN0_	167
\$_DFF_PN1_	3
\$_DFFE_PN0P_	2,041
\$_DFFE_PN1P_	2
\$_AND_	412
\$_ANDNOT_	5,057
\$_NAND_	344
\$_MUX_	2,853
\$_OR_	3,434
\$_NOR_	455
\$_ORNOT_	308

\$_NOT_	773
\$_XOR_	689
\$_XNOR_	115

Note: Area and timing (MHz) figures require mapping to a real liberty cell library. Run yosys scripts/synth_gf180.sh with the GF180MCU full liberty for area estimates.

Tool Versions

Tool	Version
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Yosys	0.58+35 (git sha1 89f32a415)
VCS	X-2025.06
Verilator	5.041
PDK	GF180MCU (globalfoundries-pdk-

09 — Requirements Traceability Matrix (RTM)

Coverage: 23 REQ-IDs total · 8 signed-off · 15 locked

Source: schemas/rtm.json + schemas/verification_report.json

REQ-ID	Description	RTL Module(s)	SVA File	TC(s)	Sim	Mut	S/O
REQ-F01	RV32I base ISA (RV32I instructions)	rv32im_core.sv	rv32im_ext.sva	TC-001..003, TC-011		N/A	✓
REQ-F02	RV32M multiply/divide instructions	rv32im_ext.sv	rv32im_ext.sva	TC-017..018	✓	N/A	✓
REQ-F03	FENCE.I instruction support	rv32im_decoder, rv32im_if_stag	TC-009	✓	N/A	✗ (no SVA)	
REQ-F04	WFI instruction support	rv32im_decoder, rv32im_if_stag	TC-019	✓	N/A	✗ (no SVA)	
REQ-F05	Configurable boot address (PMA)	rv32im_if_sva.sv	—	✗	N/A	✗ (no TC)	
REQ-F06	Load-use stall (1-cycle hazard)	rv32im_if_sva.sv	TC-010	✓	70%	✗	
REQ-F07	EX→EX / MEM→EX forwarding	rv32im_if_sva.sv	TC-002..005, TC-011	70%	✗		
REQ-F08	Branch resolution	rv32im_id_sva.sv	TC-006..008	✓	30%	✗	
REQ-F09	32×32 register file	rv32im_id_sva.sv	TC-001, TC-011	✓	N/A	✓	
REQ-F10	Load/store byte-enable sign/	rv32im_mem_sva.sv	TC-004..005	✓	N/A	✓	
REQ-F11	Machine-mode CSR link	rv32im_ex_sva.sv	TC-012..013, TC-020	N/A	✓		
REQ-F12	Synchronous exception handling	rv32im_ex_sva.sv	TC-014	✓	45%	✗	
REQ-F13	Asynchronous interrupt handling	rv32im_ex_sva.sv	TC-015..016	✓	45%	✗	
REQ-F14	Instruction fetch buffer	rv32im_mem_sva.sv	TC-021..022	✓	N/A	✓	
REQ-F15	Data memory bus error handling	rv32im_mem_sva.sv	TC-021..022	✓	N/A	✓	
REQ-F16	Bus error handling	rv32im_wb_sva.sv	TC-023	✓	N/A	✓	
REQ-F17	Performance counter	—	—	✗	N/A	✗ (no TC)	
REQ-F18	Vectored trap mode	—	—	✗	N/A	✗ (disabled)	
REQ-C01	Single clock domain	rv32im_core (top)	—	✗	N/A	✗ (no TC)	
REQ-C02	Active-low synchronous reset	rv32im_core (top)	—	✗	N/A	✗ (no TC)	
REQ-C03	No combinational logic through	rv32im_core (top)	—	✗	N/A	✗ (no TC)	
REQ-C04	PR_BUS_OUTSTANDING if (stage)	rv32im_isu	—	✗	N/A	✗ (no TC)	
REQ-C05	GF180MCU synthesis rules	—	—	✗	N/A	✗ (no TC)	

Document generated by Claude Code RTL Gen Flow — spec_pdf_generator Phase 6.
Source artifacts: schemas/final_config.json, schemas/rtm.json,
schemas/verification_report.json, schemas/selected_testplan.json,
schemas/synth_report.json, src/rtl/filelist.f, src/rtl/rv32im_core.sv